

The Development History and Evolutionary Trends of Semiconductor

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Abstract: The exponential growth of AI technology is transforming the global semiconductor industry. The significant demand for high-performance computing chips for AI is driving continuous innovation in the semiconductor industry. As the evolution of Moore's Law is slowing down, new technologies should be explored to continue Moore's Law. The development of semiconductors will become more diverse in the future. New-structure transistors, new semiconductor materials, three-dimensional integrated circuits (3DICs), and DTCO are all potential avenues for the future development of semiconductors. A study of the evolutionary trend of the semiconductor is highly valuable. This paper presents a historical analysis of the development of semiconductor processes from 1947 to the present day, with a particular focus on the structure of FinFET and GAA. This article presents three examples of DTCO for the purpose of enhancing transistor density. Based on the findings of this analysis, it can be concluded that DTCO is the most effective and cost-effective method for increasing transistor density, and that DTCO will play an important role in the future development of semiconductors.

Keywords: Transistor, evolutionary trends, semiconductor.

1. Introduction

The invention of the point-contact transistor by Bell Labs in 1947 marked the beginning of the semiconductor era. In 1958, Texas Instruments developed the world's first digital integrated circuit, which was a significant milestone in the history of integrated circuits. Subsequently, Fairchild Semiconductor proposed two key processes, namely lithography and planar processing, which built the foundation for the mass production of integrated circuits. Integrated circuits have become a fundamental component of modern information technology.

With the development of technology, the size of integrated circuits is getting smaller and smaller, the integration density is getting higher and higher, which has promoted the progress of computers, communications, and consumer electronics. Integrated circuits support the development of the world economy. The wide use of integrated circuits has changed people's lifestyles, from smartphones and tablets to smart electrical appliances and smart transportation. Integrated circuits have entered every aspect of people's lives. These electronic goods have not only improved the quality of people's life, but also changed their communication style, consumption habits, and working habits.

Semiconductor is the foundation for manufacturing integrated circuits, and the development of semiconductor has driven the development of integrated circuits. With the continuous optimization

and innovation of semiconductor technology, the integration density of integrated circuits continues to increase, the performance continues to enhance, and the power consumption continues to decrease, which promotes the development of the entire electronics industry. This paper analyzes the development history and current state of semiconductors, forecasts the development trend of semiconductors.

2. Development history of semiconductor

In 1904, British physicist John Ambrose Fleming invented the first vacuum diode.

In 1906, American engineer Lee de Forest invented the vacuum transistor making the electronic tube the ability to amplify and oscillate. This invention was the birth of the electronics industry.

On December 6, 1947, William Shockley, John Bardeen, and Walter Brattain invented the world's first germanium point-contact transistor at Bell Laboratories in Murray Hill, New Jersey, USA[1]. The transistor is one of the greatest inventions in modern history, which made electronic products such as radios, calculators, and computers smaller and cheaper, and had great significance for the development of integrated circuits.

In April 1950, Shockley made the first bipolar junction transistor - the PN junction transistor. Today most transistors are still PN junction transistors.

In 1952, the junction field effect transistor was manufactured [1].

In 1959, Atara and others invented a Metal Oxide Semiconductor.

In 1963, Frank and C.T. Sah first proposed CMOS (Complex Metal Oxide Semiconductor) technology, which connected n-MOS and p-MOS into a complementary structure. This bipolar mosfet was almost free of static current when they were in and on state, so they were suitable for logic circuits. CMOS technology has the advantages of low power, high integration density, and easy production. Since the 1970s and 1980s, it has gradually become the mainstream process for integrated circuit chip production. Until now, this technology has occupied more than 95% of the global semiconductor integrated circuit chip industry [2].

When mentioning semiconductors, the first thing that comes to mind is Moore's Law. In 1965, Gordon Moore proposed Moore's Law. Moore's Law states that when prices remain constant, the number of transistors in a chip will double approximately every 18 months, and the performance will increase twice. The development of CMOS effectively validates the accuracy of Moore's Law.

Table 1 shows the transistor densities of Intel 90nm, 65nm, 45nm, 32nm, 22nm, 14nm, 10nm, and 7nm process nodes. The data in Table 1 also validates the accuracy of Moore's Law.

Table 1: Transistor density in intel process node

Process Node	90nm	65nm	45nm	32nm	22nm	14nm	10nm	7nm
Density (MTr/mm ²)	1.45	2.08	3.33	7.11	16.5	44.67	100.76	212.48

The essence of Moore's Law is increasing transistor operating speed, increasing transistor density, and decreasing costs by reducing transistor feature size. However, as the feature size of transistors continues to decrease proportionally, the distance between the source and drain also becomes shorter. The channel is not only affected by the gate electric field but also by the source electric field. The control ability of the gate over the channel decreases, and the difficulty of the gate voltage clamping off the channel increases. This can easily result in leakage and the formation of the Short Channel Effect (SCE), which can cause serious degradation of the transistor and affect switching speed. Due to the above issues, it is difficult to further reduce the feature size of planar transistors. In 1999,

Professor Hu Zhengming from the University of California, Berkeley invented FinFET (Fin Field Effect Transistor), which is a new type of three-dimensional transistor structure. This invention pioneered a new era of MOS structures. In 2009, Intel was the first to adopt this technology, and the 22nm process node released in 2011 was the industry's first process to use FinFET transistors.

FinFET has been used as a transistor in cutting-edge semiconductors for over a decade since 2011. But By 2022, when 5nm process node was in production, FinFET also faced problem such as leakage, short channel effect, and threshold voltage drift which due to transistor size shrinking[3].In order to solve these problems and the evolution of Moore's Law, it is urgent for the emergence of new transistors.

In 1990, Dr. Cor Claeys and his research team from IMEC in Belgium proposed the GAA technology.Compared to FinFET, transistors using GAA technology have stronger gate control over the channel. Currently, GAA has become the focus of attention in the semiconductor industry.

3. Current state of semiconductor process

Table 2 lists the most advanced process nodes of Intel, Samsung, and TSMC, the three most advanced semiconductor manufacturers in the world. Table 2 shows that FinFET has become the mainstream technology in advanced process nodes today. Samsung adopted GAA technology at the 3nm process node, making it the first semiconductor company in the industry to adopt GAA technology.

Table 2: Current status of advanced technologies from Intel, Samsung, TSMC

	Intel			Samsung				TSMC			
Process Node	14nm	10nm	7nm	10nm	7nm	5nm	3nm	10nm	7nm	5nm	3nm
	FinFET	FinFET	FinFET	FinFET	FinFET	FinFET	GAA	FinFET	FinFET	FinFET	FinFET

This chapter focuses on FinFET and GAA, which are currently the two mainstream transistors.

The 3D structural schematic of FinFET is shown in Figure 1. The channel of FinFET is composed of high and thin fins protruding from the substrate, and the drain and source are located on both sides. Compared with the planar transistor, the FinFET solid fin structure enhances the gate's control ability over the channel and has significant advantages in controlling short channel effects, improving subthreshold characteristics, and suppressing gate leakage current.

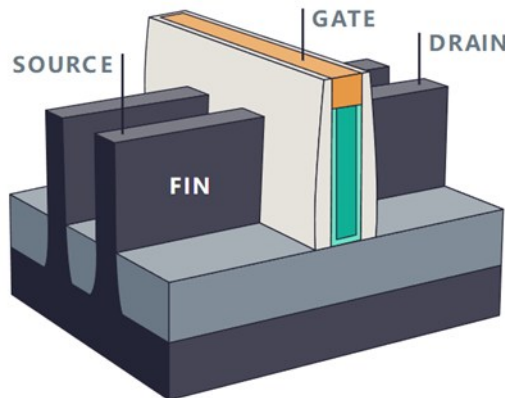


Figure 1: The schematic of FinFET transistor structure[4]

The invention of FinFET transistor is a significant technological breakthrough with high practical application value. It not only solves the leakage problem in the semiconductor process, but also promotes the continuation of Moore's Law, enabling the rapid development of electronic products.

Since 2011, mobile electronic products have entered a stage of rapid development, and smartphones are the core of mobile electronic products, providing various functions such as communication and entertainment. The development of smartphones benefits from the advancement of semiconductor technology, which enables more transistors integrated into core components such as processors and memory in smartphones, thereby improving performance. And, the advanced semiconductor technology also helps to reduce the power consumption of chips and extend the battery life of smartphones. In addition, with the continuous reduction of chip size, smartphones can achieve thinner and lighter designs. The process node used by Apple smartphone SOC chips is shown in table 3. The SOC chips of iPhone15 Pro, iPhone15 Pro Max, and iPhone16 series smartphones all use the most advanced TSMC 3nm FinFET technology

Table 3: The process node of Apple Smartphone SOC chip

Year	Phone Type	SOC Chip	Foundry	Process Node
2010	iPhone4	A4	Samsung	45nm
2011	iPhone4S	A5	Samsung	45nm
2012	iPhone5	A6	Samsung	32nm
2013	iPhone5S	A7	Samsung	28nm
2014	iPhone6	A8	TSMC	20nm
2015	iPhone6S	A9	Samsung/TSMC	14nm/16nm
2016	iPhone7	A10	TSMC	16nm+
2017	iPhoneX	A11	TSMC	10nm
2018	iPhoneXs	A12	TSMC	7nm
2019	iPhone11 Pro	A13	TSMC	7nm
2020	iPhone12 Pro	A14	TSMC	5nm
2021	iPhone13 Pro	A15	TSMC	5nm
2022	iPhone14 Pro	A16	TSMC	4nm
2023	iPhone15 Pro	A17	TSMC	3nm
2024	iPhone16 Pro	A18	TSMC	3nm

Gate-All-Around(GAA) is a gate all around the 3D transistor[5]. In this transistor, the gate completely surrounds the source and drain, forming four contact surfaces and improving the control ability of the channel.

GAA is further divided into nanowires (shown in Figure 2) and nanosheet (shown in Figure 3) due to the different channel shapes. The channel shape of nanowire is a slender linear shape, while the channel shape of nanosheet is a flat sheet shape, with a larger contact area between the gate and the channel. Because nanosheet has wider channels, they have stronger driving currents and better performance than nanowire.

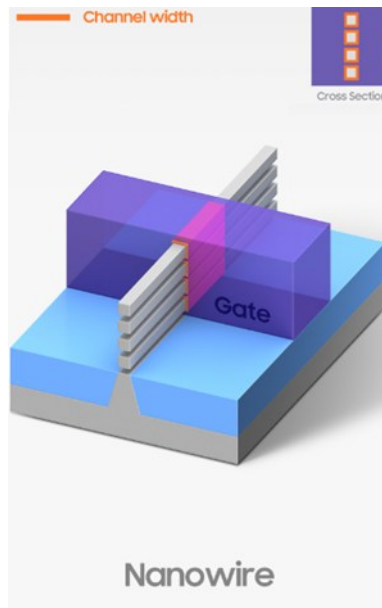


Figure 2: The schematic of nanowire structured transistor[6]

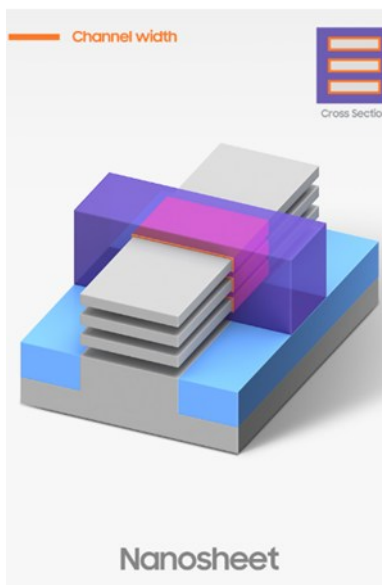


Figure 3: The schematic of nanosheet structured transistor[6]

4. The trend of the semiconductor process

Due to the reduction in transistor feature size approaching the physical limit, the cost of developing and producing smaller chips is also increasing, and the manufacturing cycle time is also getting longer. The industry and academia are exploring new solution to address the slowdown of Moore's Law. Now there are three solutions that have made progress. The first solution is the new transistor technologies, such as Forksheet [7] and Complementary-FET[8], which are currently under research. The second solution is DTCO (Design Technology Co optimization), which improves the density of transistors per unit area without changing the feature size of the transistors. The third solution is 3DIC, which achieves the goal of integrating more transistors in a smaller space by stacking multiple dies together.

DTCO is a relatively mature technology, such as TSMC achieving an increase in transistor density through DTCO at multiple advanced process nodes. This chapter focuses on DTCO technology.

The definition of DTCO is quite broad, and any co-optimization measures between the semiconductor process and specific circuit designs can be referred to as DTCO. Generally speaking, conventional semiconductor chip design involves optimizing circuit design based on the given process and PDK(process design kit). Therefore, in the context of recent years, DTCO mainly refers to how to optimize semiconductor process according to the specific requirements of semiconductor chips. Specific process optimization can include materials for metal interconnects, via design, specific CMOS device design, and so on.

In the golden age of Moore's Law, as a new generation of semiconductor processes emerges every year with significant performance improvements, the benefits of spending a lot of time and resources on DTCO optimization for this generation of processes are not significant. However the continuation of Moore's Law is becoming more and more challenging, and the cost of developing and using new-generation semiconductor process is getting higher. At the same time, the performance improvement brought by the new-generation semiconductor process is getting smaller and smaller. It is more practical to use DTCO to optimize semiconductor processes and circuit design.

Figure 4 shows the proportion of DTCO contribution to transistor density increasing in TSMC at 16nm, 10nm, 7nm, 5nm, and 3nm process nodes.

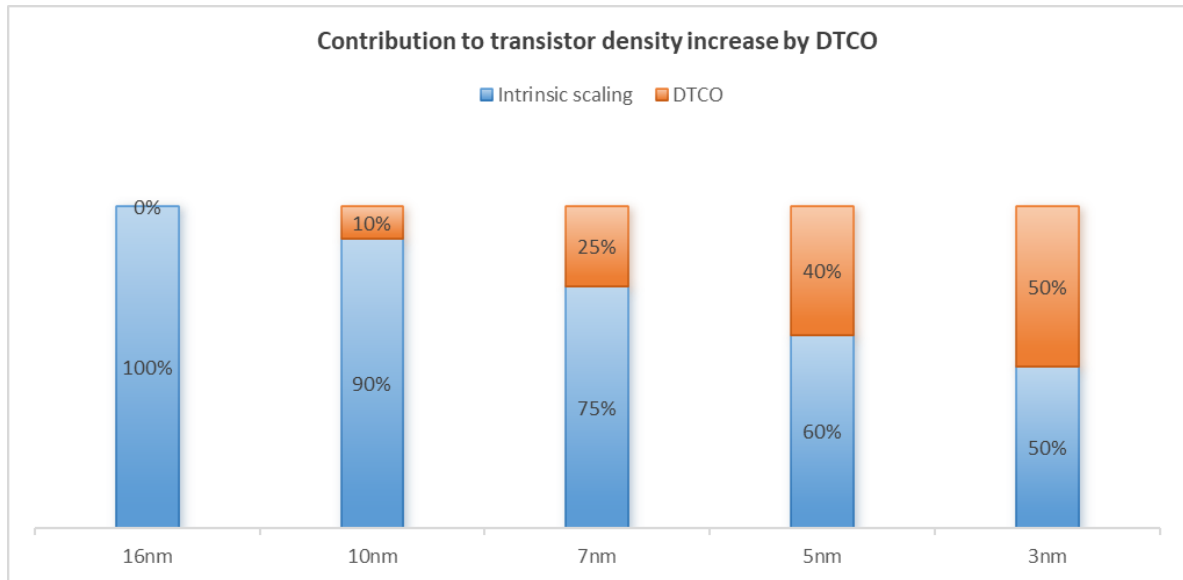


Figure 4: Contribution to transistor density increase by DTCO[9]

Now, DTCO plays a crucial role in the semiconductor process and is one of the key technologies driving the continuous progress and development of the semiconductor industry. Here are a few examples of using DTCO technology to increase the density of unit transistors[10].

4.1. DDB to SDB

Diffusion Break(DB) is used in a standard cell to isolate two adjacent devices, prevent diffusion currents between adjacent devices from affecting each other, and ensure the stability and stability of the devices.

Double Difufsion Break(DDB)is a design where adjacent devices have independent Shllow Trench Isolation(STI), providing good isolation while also increasing the width of standard cells, as shown in Figure 5.

Single Diffusion Break (SDB), where adjacent devices share STI, reduces the standard cell width to increase transistor density, as shown in Figure 6.

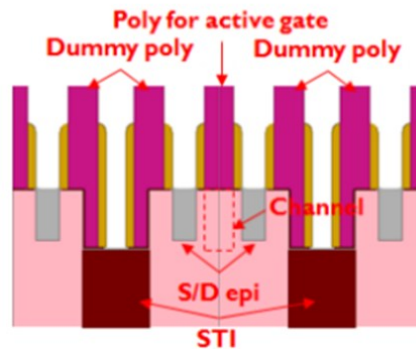


Figure 5: DDB[11]

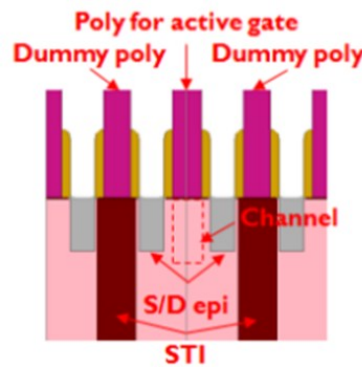


Figure 6: SDB[11]

4.2. COAG

COAG technology is that gate contact point is the control point for the source drain current in a transistor. Figure 7 is a traditional top view of a transistor. Figure 8 is a top view of a transistor using COAG technology. Placing the "gate contact" directly above the "active" area of the transistor can save space and enable the integration of more transistors in the same area. Intel 10nm process node used COAG technology, resulting in a reduction of approximately 10% in area. Due to the close contact between the gate and the contact point, COAG technology can improve the performance of transistors, including response speed and stability.

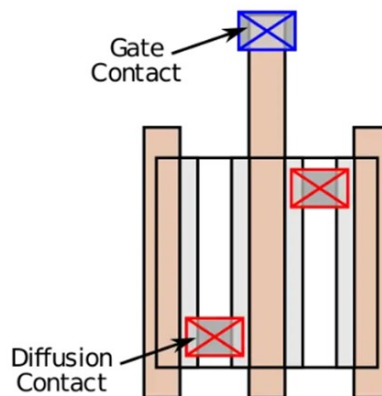


Figure 7: Top view of traditional transistor[12]

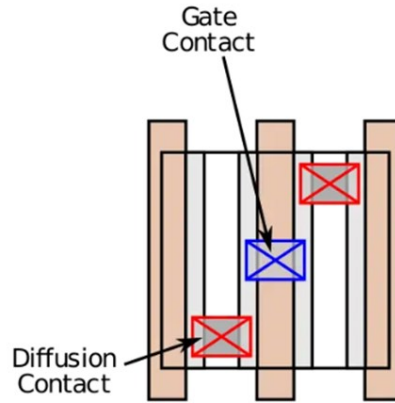


Figure 8: Top view of transistor using COAG technology [12].

Although COAG can bring benefits in terms of area and performance, it also faces some challenges. Firstly, the implementation of COAG technology requires precise process control, including photolithography, etching, and ion implantation, which increases the difficulty and cost of manufacturing; Secondly, due to the close contact between the gate and the contact point, COAG technology may lead to some manufacturing difficulties and yield issues that require further optimization and improvement.

4.3. Fin depopulation

Fin depopulation technology reduces the height of transistors by decreasing the number of Fins, allowing for the integration of more transistors per unit area. This technology can reduce power consumption and increase transistor density by reducing the number of Fins without sacrificing too much performance. Figure 9 is a schematic diagram of reducing the transistor height by decreasing the number of Fins inside the transistor.

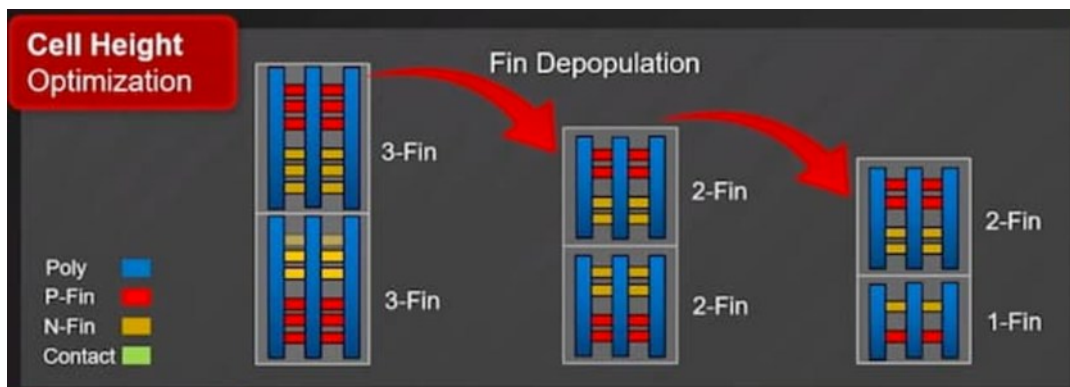


Figure 9: Schematic diagram of reducing the transistor height by decreasing Fin number[8]

Although Fin depopulation technology has the above advantages, it also faces some challenges in the manufacturing process. Firstly, reducing the number of Fins and lowering the height of transistors requires higher process control accuracy to ensure the performance and reliability of transistors. Secondly, Fin depopulation technology requires more complex designs to ensure the integrity of connections and signal transmission between transistors.

5. Conclusion

From the development of semiconductor process, it can be seen that from invention to mass production of new structured transistors takes several years, which involves multiple complex tasks, including research and development, testing, optimization, and production line preparation. Each task requires a significant investment of time, manpower, and material resources. The development of new structure transistors such as Forksheet and Complementary-FET is still in the research stage, and the time for mass production is uncertain. In order to ensure the continued evolution of Moore's Law, it is essential to pursue the development of new structure transistors while also exploring alternative avenues, such as the utilisation of novel materials and 3DIC to enhance the density of unit space transistors. The preceding analysis has identified DTCO as the most advanced technology for augmenting transistor density and performance, with extensive deployment across multiple process nodes, thereby offering promising avenues for the advancement of semiconductor processing.

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